

Write a program to implement Alpha – Beta pruning for tree and TIC
– TAC - TOE game

Program Code:

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# program to demonstrate
# working of Alpha-Beta Pruning
# Initial values of Alpha and Beta
MAX, MIN = 1000, -1000
# Returns optimal value for current player
#(Initially called for root and maximizer)
def minimax(depth, nodeIndex, maximizingPlayer,
            values, alpha, beta):
    # Terminating condition. i.e
    # leaf node is reached
    if depth == 3:
        return values[nodeIndex]
    if maximizingPlayer:
        best = MIN
        # Recur for left and right children
        for i in range(0, 2):
            val = minimax(depth + 1, nodeIndex * 2 + i,
                          False, values, alpha, beta)
            best = max(best, val)
            alpha = max(alpha, best)
            # Alpha Beta Pruning
            if beta <= alpha:
                break
        return best
    else:
        best = MAX
        # Recur for left and
        # right children
        for i in range(0, 2):
            val = minimax(depth + 1, nodeIndex * 2 + i, True, values,
                          alpha, beta)
            best = min(best, val)
            beta = min(beta, best)
            # Alpha Beta Pruning
            if beta <= alpha:
```

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        break
    return best
# Driver Code
if __name__ == "__main__":
    values = [3, 5, 6, 9, 1, 2, 0, -1]
    print("The optimal value is :", minimax(0, 0, True, values, MIN,
MAX))
```